INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Infosys SE Complete Preparation — LCM and HCF Axioms & Applications

Section: prepinsta infosys SE | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Infosys SE Complete Preparation — LCM and HCF Axioms & Applications

Source Portal: PrepInsta Infosys Preparation Suite (`https://prepinsta.com/infosys/aptitude/lcm-and-hcf/`)
Target Role: Infosys Systems Engineer (SE) | **Category:** 01_Mathematical_Ability_Technical_Aptitude
Total Verified Questions Extracted: 10
Quizzes Covered: quiz-1, quiz-2, quiz-3, quiz-4

📌 Section 1: Rules & Core Concepts to Keep in Mind

These rules and fundamental principles govern all Infosys questions on this topic:

- Product of two numbers = LCM * HCF.
- HCF of fractions = HCF of Numerators / LCM of Denominators.
- LCM of fractions = LCM of Numerators / HCF of Denominators.
- HCF of (a, b) always divides LCM of (a, b).
- HCF and LCM

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

  - Number System

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

- Number Decimals & Fractions

```
Questions

Formulas

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```

  - Surds and Indices

```
Questions

Formulas

How to Solve Quickly

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```

    - Divisibility

```
Questions

Formulas

How to Solve Quickly

Shortcuts, Tips & Tricks
```

      - Ages

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

        - LCM

```
Questions

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- HCF

```
Questions

Formulas

How To Solve Quickly

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```

  - Inverse

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

    - Speed Time and Distance

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

      - Work and Time

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

- Boats and Streams

```
Questions

Formulas

How To Solve Quickly

Shortcut, tips, and tricks
```

⚡ Section 2: How to Solve Quickly (Exam Elimination Strategies)

Proven time-saving strategies recommended by PrepInsta for Infosys 2026 examination:

1. **Option Elimination via Unit Digits & Multiples:** Before doing full calculations, inspect the options to see if last digits or divisibility by 3, 7, 9, 11 can eliminate 2 to 3 wrong choices.
2. **Approximation & Magnitude Checks:** When exact arithmetic is tedious, round numbers to neat multiples (e.g., 99 ≈ 100) to find the nearest valid option.
3. **Reverse Option Substitution:** In complex algebraic equations, plug option values directly into the question stem to test validity.

🎯 Section 3: High-Yield Tricks & Shortcuts

Key mathematical formulas, mental-math shortcuts, and speed axioms:

- Greatest number dividing x, y, z leaving remainder r in each case = HCF(|x-y|, |y-z|, |z-x|).
- Smallest number which when divided by x, y, z leaves remainder r in each case = LCM(x, y, z) + r.
- HCF and LCM

```
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  - HCF

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      - Speed Time and Distance

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📝 Section 4: Comprehensive PrepInsta Practice Question Bank (All Quizzes)

Contains **10** verbatim questions with options, verified answer keys, and complete derivations:

Question 1 (PrepInsta quiz-1 - Q1)

Question Stem:
Find the GCD of any 3 consecutive numbers.

Options:
- **(A)** `1`
- **(B)** `2`
- **(C)** `3`
- **(D)** `4`

Correct Answer: (A) `1`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Solution: Let the three numbers be 7, 8, 9




Factors of 7 are = 1 , 7




Factors of 8 are = 1, 2, 4, 8




Factors of 9 are = 1, 3, 3, 9




We can see that the Greatest common factor is 1.
```

Question 2 (PrepInsta quiz-1 - Q2)

Question Stem:
Find LCM of 12, 15, 20, 27.

Options:
- **(A)** `500`
- **(B)** `540`
- **(C)** `550`
- **(D)** `None of these`

Correct Answer: (B) `540`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Sol: 12 = 2×2×3 = 2^2 ×3^1



15 = 3× 5



20 = 2×2×5 = 2^2×5



27 = 3×3×3 = 3^3



 



HCF = 1



LCM = 2^2 × 3^3 × 5 = 540
```

Question 3 (PrepInsta quiz-1 - Q3)

Question Stem:

There are two numbers. HCF of both the numbers is 11, and their LCM is 693. If the first number is 77, find the second number?

Options:

- **(A)** `89`
- **(B)** `56`
- **(C)** `78`
- **(D)** `99`

Correct Answer: (D) `99`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
The product of two numbers = HCF * LCM



Let the required number is = x



So, 77 * x = 11 * 693



Hence, x=99.
```

Question 4 (PrepInsta quiz-1 - Q4)

Question Stem:

Find the greatest number which divides 34, 90, and 104 and leaves the same remainder in each case.

Options:

- **(A)** `15`
- **(B)** `17`
- **(C)** `14`
- **(D)** `18`

Correct Answer: (C) `14`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Difference between numbers = 90-34 = 56, and 104- 90=14



HCF of 56 and 14 = 14



Hence, 14 is the largest number which divides the given number and leaves the same remainder in
each case.
```

Question 5 (PrepInsta quiz-1 - Q5)

Question Stem:
What is the least multiple of 7 which when divided by 6, 9, 15, and 18 respectively, leaves 4 as the remainder in each case?

Options:
- **(A)** `364`
- **(B)** `255`
- **(C)** `385`
- **(D)** `266`

Correct Answer: (A) `364`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Take LCM of 6, 9, 15, and 18. Their LCM = 90


As per given information, (90* x+4)/7




(90 *x)/ 7 = ((7 * 12 *x) + 6x + 4)/7

Or, (7*12x)/7 + (6x + 4)/7




x=4 satisfy the condition.

Hence, the required number is 90 * 4 + 4 = 364.
```

Question 6 (PrepInsta quiz-1 - Q6)

Question Stem:
Which least number should be subtracted from 1936 so that the resulting number, when divided by 9, 10, and 15, will leave the same remainder 8 in each case?

Options:
- **(A)** `39`
- **(B)** `38`
- **(C)** `37`
- **(D)** `36`

Correct Answer: (B) `38`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Take LCM of 9, 10, and 15. Their LCM is = 90



If the number 1936 is divided by 90, we get the remainder 46, but we need 8 as remainder.



Hence, the required number is 46 - 8 = 38.
```

Question 7 (PrepInsta quiz-1 - Q7)

Question Stem:
Find the least number which when divided by 5, 6, 7, and 8 leaves remainder 3, but when divided by 9 leaves remainder 0.

Options:
- **(A)** `843`
- **(B)** `1683`
- **(C)** `2523`
- **(D)** `3363`

Correct Answer: (B) `1683`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Take LCM of 5, 6, 7, and 8. It will be = 840.



Let the number 840*x + 3 is completely divisible by 9 if the sum of digits is divisible by 9.



Let x= 2, the number will be 1683, and the sum of its digits is equals to 18 which is divisible by
9. So, the required number is 1683.
```

Question 8 (PrepInsta quiz-1 - Q8)

Question Stem:
What is the number nearest to 10000 which is exactly divisible by 3, 4, 5, 6, 7, and 8?

Options:

- **(A)** `9956`
- **(B)** `10080`
- **(C)** `10096`
- **(D)** `9240`

Correct Answer: (B) `10080`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Take LCM of 3, 4, 5, 6, 7, and 8. It will be = 840.



Now, 10000/840 = 760 (remainder).



840 - 760 = 80. If we add 80 in the given number, the number is exactly divisible by 840.



So, the required number is 10080.
```

Question 9 (PrepInsta quiz-1 - Q9)

Question Stem:

The LCM of two prime number x and y is 161. If x>y, find the value of 15y-7x.

Options:

- **(A)** `-32`
- **(B)** `-56`
- **(C)** `38`
- **(D)** `74`

Correct Answer: (B) `-56`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
Note: The HCF of two prime numbers = 1



LCM/HCF =161/1 =161

161=7 * 23

The ratio of numbers: 7: 23

Hence, the expression gives15 * 7 - 7 * 23 = -56
```

Question 10 (PrepInsta quiz-1 - Q10)

Question Stem:

Find the smallest digit that will leave same remainder 4 in each case when divided by 12, 15, 20, and 54.

Options:

- **(A)** `454`
- **(B)** `540`
- **(C)** `544`
- **(D)** `450`

Correct Answer: (B) `540`

PrepInsta Detailed Explanation & Mathematical Derivation:

```
 



Explanation:



Take LCM of 12, 15, 20, and 54. It will be = 540.



ATQ, the remainder in each case = 4.



Hence, the required number = 540+4 = 544.
```